Table 3. Predicted and Actual Density of DES Modeled in COSMO at 25°C and 1 atm

Solvent HBA:HBD	Predicted Solvent Density (g/L)	Reference Solvent Density (g/L)
Glycol	1,057.8	1110 [66]
Glycerol	1,176.2	1260 [66]
Cyclohexane	769.4	779
N_{4444}Br:Glycol 1:2	1,038.8	1,066.6 [67]*
N_{6666}Br:Glycol 1:2	996.7	1,007.8 [21]
ChCl:Glycol 1:2	1,123.1	1,117.4 [68]
METPB:Glycol 1:3	1,231.4	1,250.1 [68]
N_{4444}Br:Glycerol 1:2	1,079.9	1,135.6 [67]*
N_{6666}Br:Glycerol 1:2	1,031.6	1,045.8 [21]
ChCl:Glycerol 1:2	1,181.4	1,192.0 [13]
METPB:Glycerol 1:3	1,266.6	1,297.6 [13]

*Density Referenced at 30°C

Almost of the combined DES and pure compounds taken from COSMOBase have predicted densities lower than expected, with the exception of ChCl:Glycol. The most severe of these is glycerol, which is nearly 6.7% lower than the expected density of 1,260 g/L at 1,176. Glycol is predicted to be 4.7% less dense than expected at 1,057 g/L compared to 1,110. N4444Br:Glycerol also has a density that is 4.9% lower than expected at 1,079 g/L compared to 1,135.[67] All other DES and pure compounds are within 2.6% of the experimentally determined densities. These deviations, being consistently lower than expected, could be due to either bond lengths in the individual molecules being too long or due to intermolecular forces being predicted as too low. Too weak of attractive forces would cause more space between molecules as repulsive electrostatic forces between similarly charged surfaces push molecules apart.

2.3 Modeled DES Results; Electron Density

One of the main outputs of the COSMO model is the electron density, also known as sigma or σ, plot of the molecules involved in the model. These plots are referred to as sigma profiles where the area of sigma, ρ(μ), is plotted versus electron density (μ). The total area under the curve of these profiles is the total area of the molecule. Figure 15 shows the sigma profiles of pure solvents sulfolane, ethylene glycol, and glycerol as well as cyclohexane and benzene. These profiles are what are used by COSMO to complete calculations, such as the ones used to predict solubility in Table 4. These profiles can also be used with a basic understanding intermolecular forces to qualitatively assess the driving forces behind desirable solvent-solute interactions.

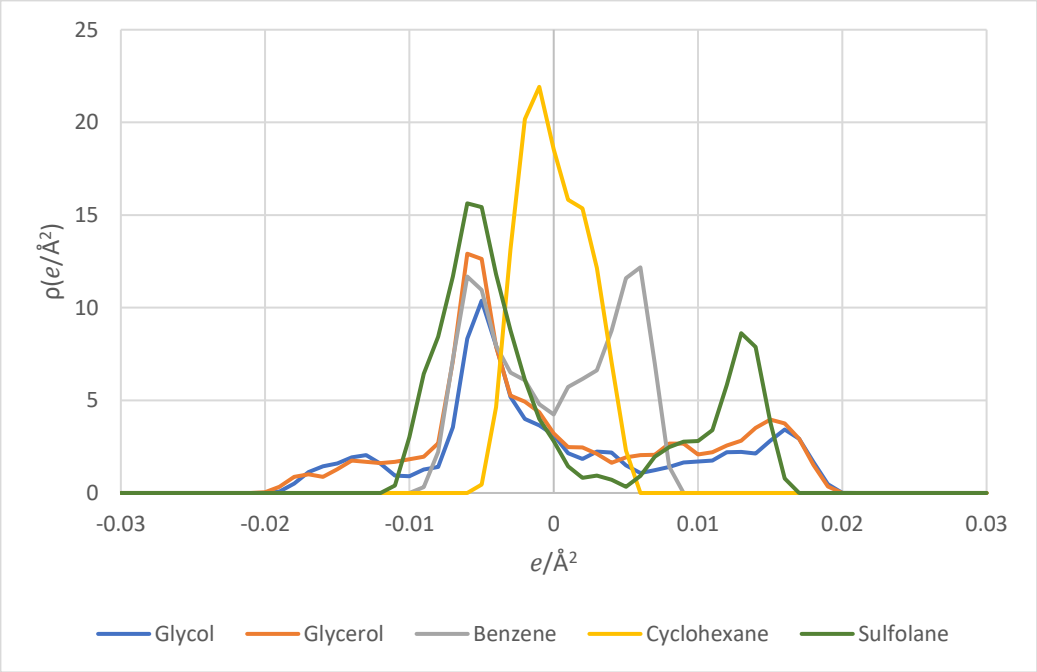


Figure 15. Sigma Profiles of Pure Solvents and Solutes

Benzene has two peaks where the electron rich peak at 0.006 $\frac{e}{Å^2}$ belongs to the delocalized aromatic π bonds and the electron deficient peak at -0.006 $\frac{e}{Å^2}$ belongs to the hydrogens. Please note that the sigma profile plot does not sum to zero across all density values. It is not uncommon for there to be more surface area on either the electron rich or deficient side. A solvent can be expected to favorably interact with benzene if it has corresponding peaks that are electron rich or deficient. Sulfolane, for example, has two distinct peaks at -0.006 and 0.013 $\frac{e}{Å^2}$. The first peak is electron deficient show the hydrogens that line the cyclic carbons. The electron rich peak is due to the sulfur and oxygen. Notably sulfolane does not have significant surface

area that is charge neutral. CO_2 occupies the electron rich sulfur-oxygen area of sulfolane when absorbed, while the electron deficient ring portion of sulfolane interacts with adjacent sulfolane.[69] Appendix A contains a related study on furfural removal from water that explains these concepts even further.

Ethylene glycol and glycerol have very similar sigma profiles as shown in Figure 15. Despite being the molar majority compound in all of the DES involved in this study, the HBA dominates the sigma profiles of the DES. Figure 16 and Figure 17 show the sigma profiles of ethylene glycol and glycerol based DES respectively. For the quaternary ammonium bromide salts $N_{4444}Br$ and $N_{6666}Br$, the non-polar arms of the salts overshadow the polar portions of their corresponding HBD in areas that are electron deficient. $N_{4444}Br$:EG and $N_{4444}Br$:Glycerol have electron deficient peaks at -0.004 and -0.002 $\frac{e}{Å^2}$ respectively, well below the hydrogen bonding donating threshold of -0.0081 $\frac{e}{Å^2}$.[47,50] The sharp, electron rich peak in all DES is due to the halide, which is the most electron dense portion of any of the DES. This peak is located at 0.017 $\frac{e}{Å^2}$ for bromide and 0.019 $\frac{e}{Å^2}$ for chloride containing DES.

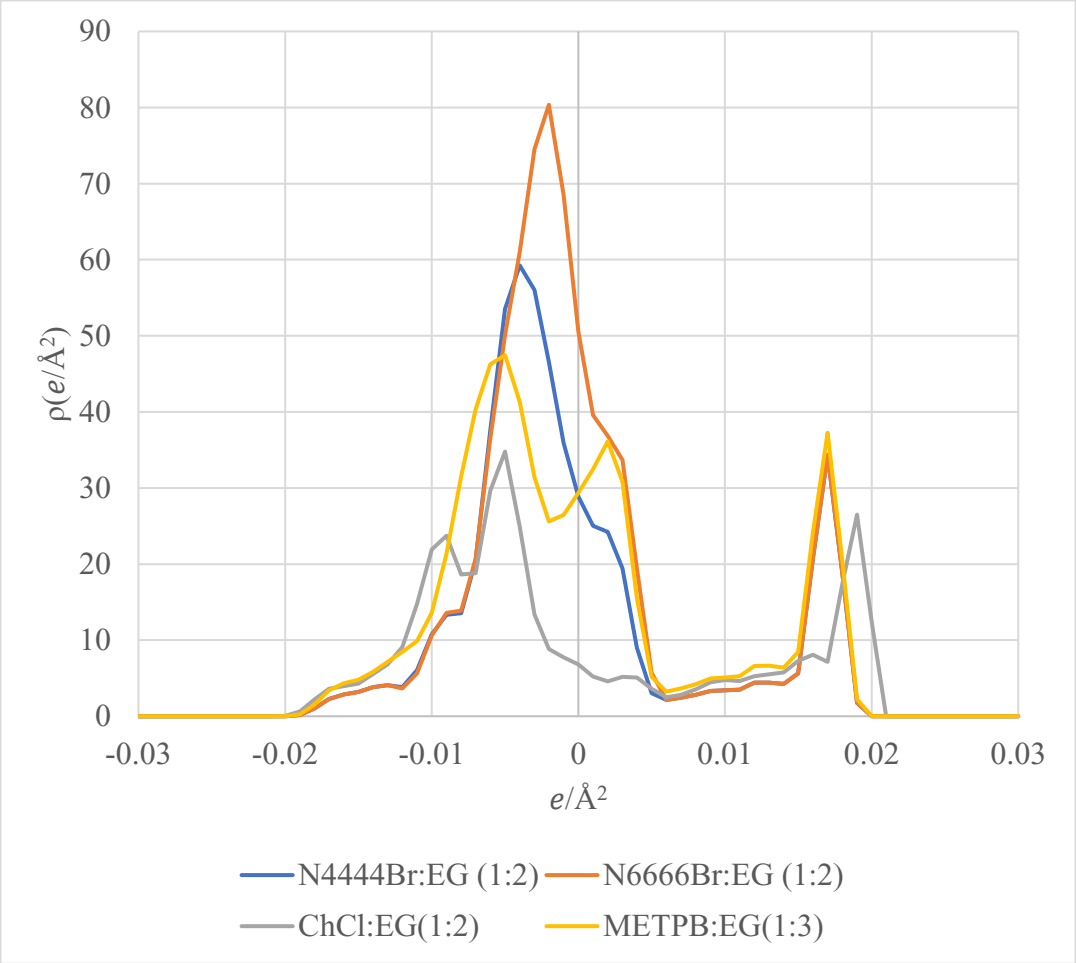


Figure 16. Sigma Profiles of Ethylene Glycol Based DES

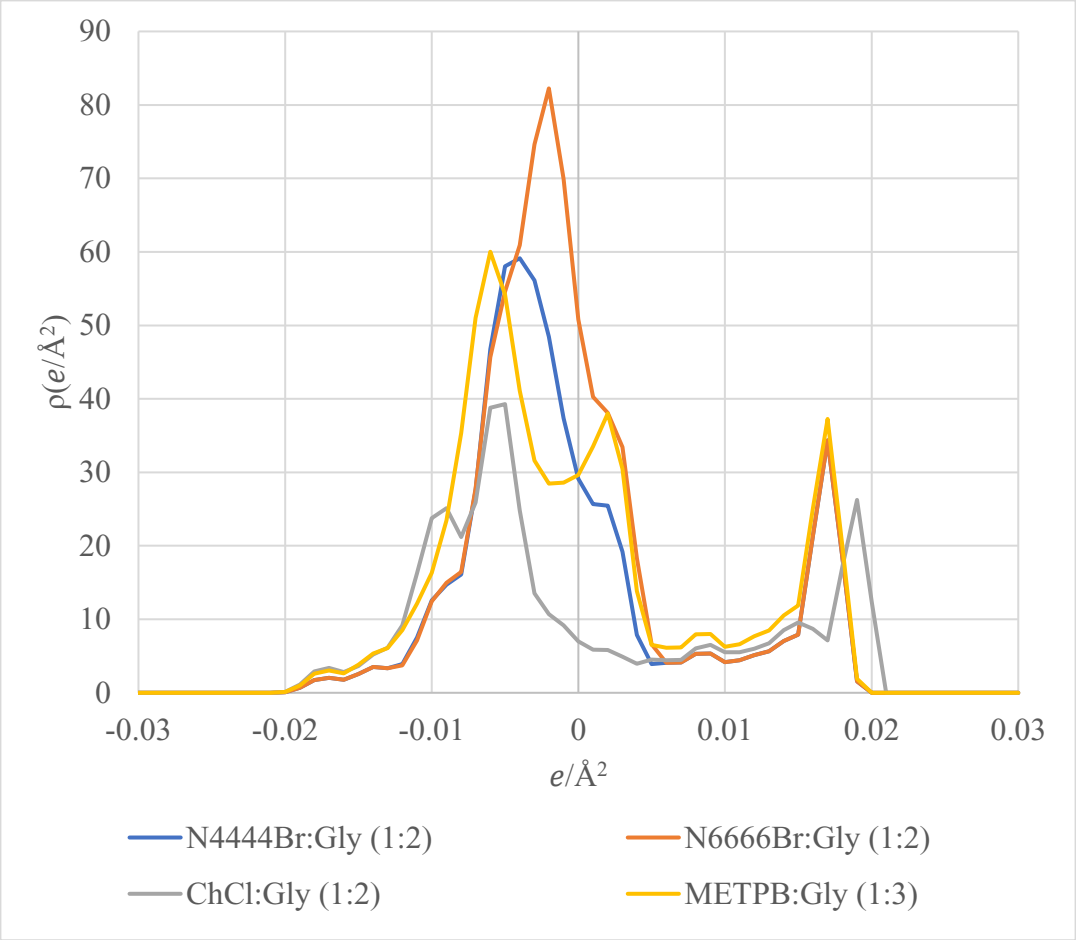


Figure 17. Sigma Profiles of Glycerol Based DES

METPB based DES have an additional peak at 0.002 $\frac{e}{Å^2}$ when paired with both glycerol and ethylene glycol. This peak is due to π bonds in the phenyl functions groups, which notably are less electron rich than benzene itself. This may be advantageous for promoting HBD-benzene interactions, as the phenyl groups are less electronegative than benzene. Choline Chloride containing DES have the most electronegative peak occurring around -0.009 $\frac{e}{Å^2}$,just beyond the hydrogen bonding

threshold. Though the nitrogen atom already has 4 function groups and is positively charged, this shows how uncovered it is compared to the nitrogen atom in either $N_{4444}Br$ or $N_{6666}Br$. Choline chloride also has the lowest charge neutral characteristic of any of the DES. Given the expectedly large contribution of Van Der Waals interactions to benzene solubility this may inhibit benzene solubility compared to the other DES.[56]

Along with the electron density profiles presented, another key output from COSMO is the sigma potential, or μ(σ), profile which shows how favorable an interaction is between a molecule and a surface of a specific charge. A key difference between the sigma profile and sigma potential profile is that while the X-axis in both cases is $\frac{e}{Å^2}$ in the sigma profile this value is attributed to the molecule itself and in the sigma potential profile this value is attributed to a surface that is interacting with the molecule. These profiles can usually be considered mirrors of one another. A larger surface area in the electron deficient region on a sigma profile, the more favorable an interaction that is predicted on the electron rich region of a sigma potential profile. This is exactly what is seen with glycerol and ethylene glycol in Figure 18. The electron deficient regions shown in Figure 15 for these species are due to hydrogens that are available for hydrogen bonding. These corresponds with a negative potential when interacting with electron rich surfaces, which would those that can accept hydrogen bonds. Though these thresholds for charges, above and below ±0.0081 $\frac{e}{Å^2}$,

are typically framed by considering hydrogen bonding it is important to consider when hydrogen bonding isn't available. For example, sulfolane has a large surface area that is electron deficient due to the strong electron withdrawing effects of the SO_2 group, but this area can not contribute to hydrogen bonding as there are no eligible hydrogens attached to either a halide, oxygen, or nitrogen.

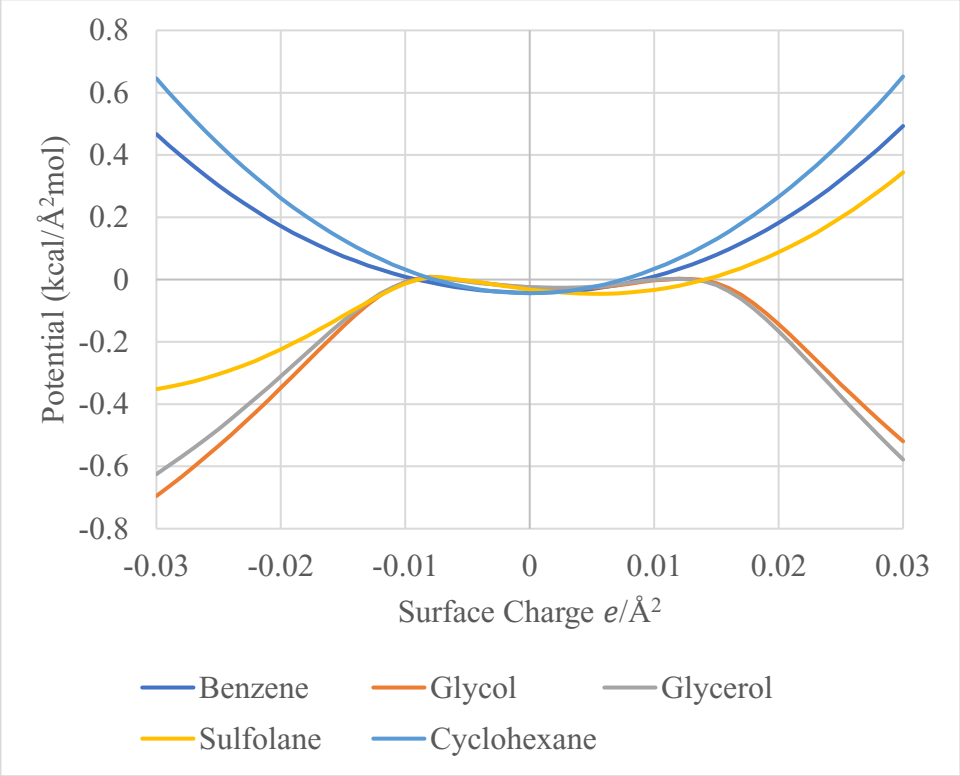


Figure 18. Sigma Potential of Pure Solvents and Benzene

The sigma potential profiles, μ(σ), for pure solvents are shown in Figure 18. There are three basic types of profiles. Benzene and cyclohexane favor interactions with neutrally charged surfaces, with the minimum potential occurring around 0 $\frac{e}{Å^2}$.

Sulfolane has a local minimum between the 0 and the hydrogen bonding threshold around 0.081 $\frac{e}{Å^2}$, showing a combination of favorable interactions with both neutrally charged surfaces and those with slight electron density. Sulfolane also has an increasingly favorable interaction displaying as a global minimum with electron deficient surfaces that can interact with the electron rich SO_2 group. Glycol and glycerol both have favorable interactions with electron rich and deficient surfaces beyond ±0.081$\frac{e}{Å^2}$ and less favorable, but still below a potential of 0 $\frac{kcal}{mol \times Å^2}$, interactions with neutral surfaces. All of the DES have similar profiles to their respective HBD's, as shown in Figure 19 and Figure 20 for ethylene glycol and glycerol respectively. One important distinction between the pure HBD and the DES is that all DES have a peak, unfavorable potentials around the electron deficient hydrogen bonding threshold of -0.081$\frac{e}{Å^2}$ and electron rich surfaces with approximately 0.015-0.017 $\frac{e}{Å^2}$. The key point for these DES is that hydrogen bonding with additional solutes is less favorable than strong electrostatic interactions as the hydrogen bonding is what is allowing the DES to be a stable liquid at room temperature.[18] Appendix A has additional information on the differences in sigma potential profiles of pure compounds and DES.

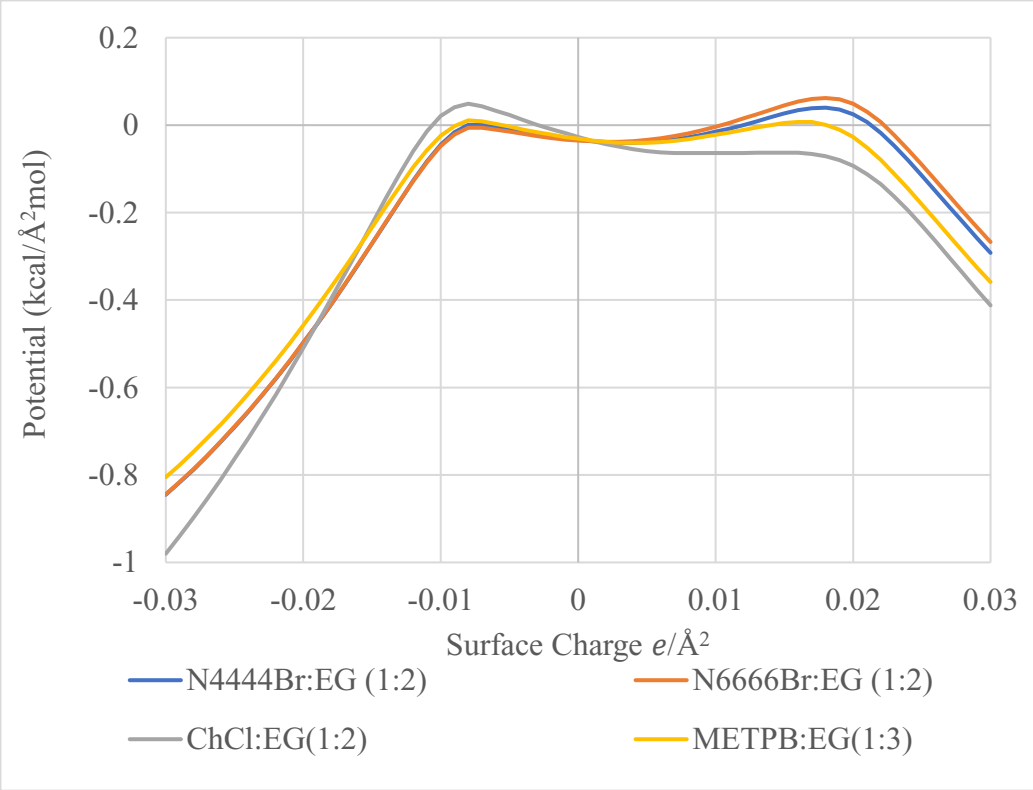


Figure 19. Sigma Potential of Ethylene Glycol Based DES

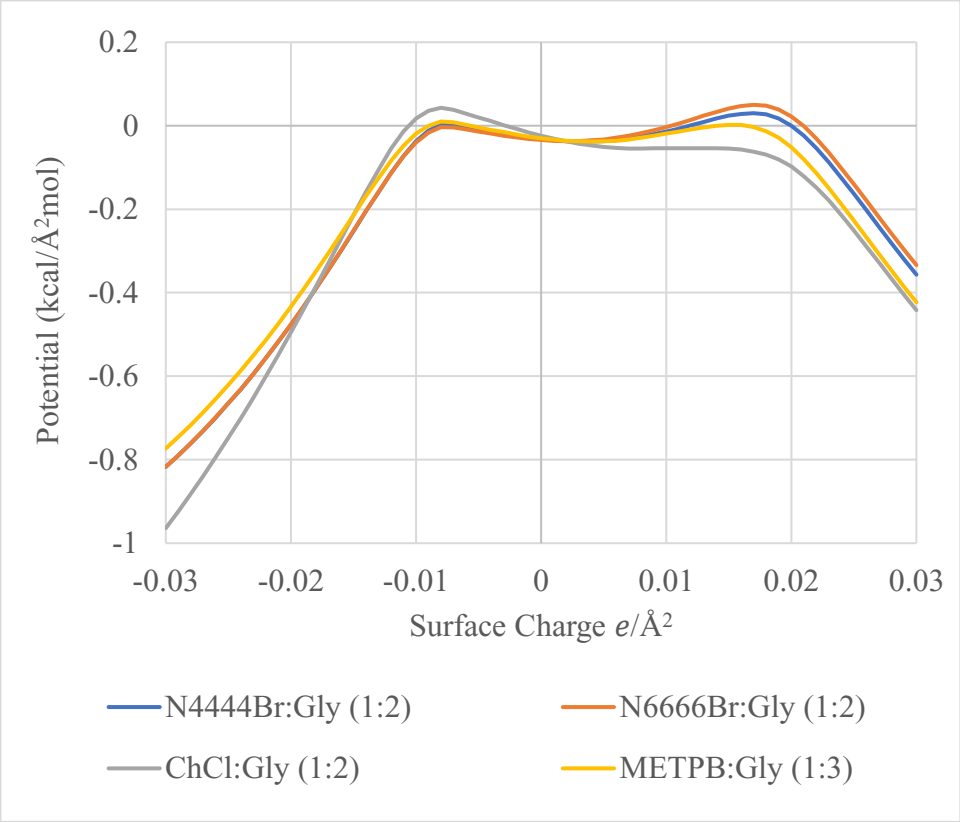


Figure 20. Sigma Potential of Glycerol Based DES

2.4 Modeled DES Results; Solubility

Determining the maximum solubility of cyclohexane and BTX species can be an efficient way to screen out DES that have either too high an affinity for cyclohexane (or other naphtha compounds) or too low an affinity for aromatic species. Table 4 shows the predicted solubilities of cyclohexane and BTX species in some selected DES as well as glycol and glycerol. The general trend from Table 4 is that cyclohexane is the least soluble compound in the DES. The aromatics species are

more soluble, but with decreasing solubility from benzene to toluene and with xylene having the lowest solubility of the aromatics. Generally, the COSMO model seems to overpredict the solubility of the aromatics in both pure solvents and in DES.

For pure ethylene glycol cyclohexane solubility is predicted to be 0.03 g/g solvent. This value is fairly close to experimental measurements from literature that determined cyclohexane solubility to be 0.02-0.04 g/g solvent.[70] Similarly, the predicted solubility of benzene at 0.17 g/g solvent is in range of experimentally determined values of 0.17-0.2 g/g solvent.[70] At 30°C ethylene glycol has a solubility of 0.035 g/g solvent for toluene, which is significantly lower than the predicted 0.11 g/g solvent (predicted at 25°C).[71]

Cyclohexane solubility in ChCl:Glycol was predicted to be 0.03 g/g solvent, while benzene was far more soluble at 0.13 g/g solvent. Cyclohexane solubility is slightly higher than the 0.02g/g solvent that is reported for ChCl:Glycol at a 1:5 molar ratio, but benzene is far higher than the 0.045 g/g solvent for the same DES.[72] Usman et al. measured the solubility of benzene in ChCl:Glycol at a 1:2 ratio at 30°C and reported 0.06 g/g solvent, which is still significantly lower than the predicted 0.13 g/g solvent.[73] Notably the solubility of all aromatic ChCl containing DES is lower than the pure solvents. Glycerol has a predicted solubility of 0.11, 0.07, and 0.04 g/g solvent for benzene, toluene, and xylene respectively. ChCl:Glycerol has lower solubilities at 0.08, 0.05 and 0.03 g/g solvent for benzene, toluene, and xylene

respectively. The predicted solubility of benzene at 25°C is close to the measure solubility of 0.09 g/g solvent for ChCl:Glycerol at 30°C.[73]

N6666Br based DES, with the increased amount of alkyl nature due to the long 6 carbon chains, had far higher solubility for all compounds than the pure solvents. N6666Br:Glycol and N6666Br:Glycerol had maximum solubilities of 0.17 and 0.13 g/g solvent for cyclohexane, approximately 6 times higher than their respective pure HBD's. Benzene solubility in N6666Br:Glycerol has been measured at 0.7 g/g solvent, while the COSMO model predicted fully miscibility.[21] N6666Br:Glycol was also predicted to have full miscibility with benzene, which is consistent with literature observations.[21]

Table 4. Predicted Solubility of Cyclohexane and BTX in Selected DES and Reference Solvents at 25°C. Solubilities are shown in $\frac{g\ Solute}{g\ Solvent}$*. Literature Experimental Values Shown in Paratheses.*

Solvent	Cyclohexane	Benzene	Toluene	Xylene
Glycol	0.03 (0.02-0.04)[70]	0.17 (0.17-0.2)[70]	0.11 (0.035)[71]*	0.07
Glycerol	0.02	0.11	0.07	0.04
N_{4444}Br:Glycol 1:2	0.11	MISC	0.52	0.33
N_{6666}Br:Glycol 1:2	0.17	MISC (0.7)[21]	MISC	MISC
ChCl:Glycol 1:2	0.03	0.13 (0.06)[73]*	0.09	0.07
METPB:Glycol 1:3	0.05	0.35	0.21	0.14
N_{4444}Br:Glycerol 1:2	0.08	0.45	0.28	0.20
N_{6666}Br:Glycerol 1:2	0.13	MISC (MISC) [21]	0.45	0.31
ChCl:Glycerol 1:2	0.02	0.08 (0.09)[73]*	0.05	0.03
METPB:Glycerol 1:3	0.03	0.22	0.14	0.09

*Referenced at 30°C

In addition to predicting the maximum solubility of BTX in each solvent, COSMOtherm was also used to generated Liquid-Liquid Equilibrium (LLE) diagrams for each solvent. All ternary diagrams are shown in Figure 21 and Figure 28. In general, the predicted LLE diagrams have either 1 or 2 phase separations. Figure 21 shows LLE between N_{6666}Br:EG, cyclohexane, and benzene which only has phase separation between the DES and cyclohexane. The DES in this case is miscible with benzene and has a phase separation with cyclohexane after 17 mass% of cyclohexane in the DES phase. Figure 22 shows a DES LLE diagram with 2 phase

separations between both METPB:Gly and cyclohexane and METB:Gly and benzene. These diagrams will be particularly useful when determining phase separations after mixing when modeling the extraction of BTX from cyclohexane in chapter 4.

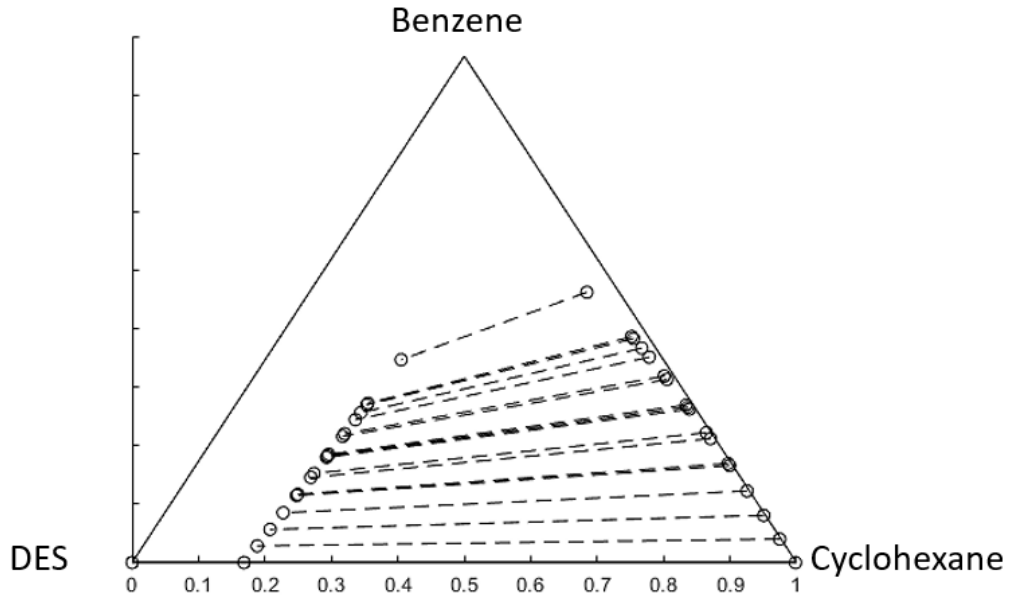


Figure 21. Predicted LLE of N6666Br:EG (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

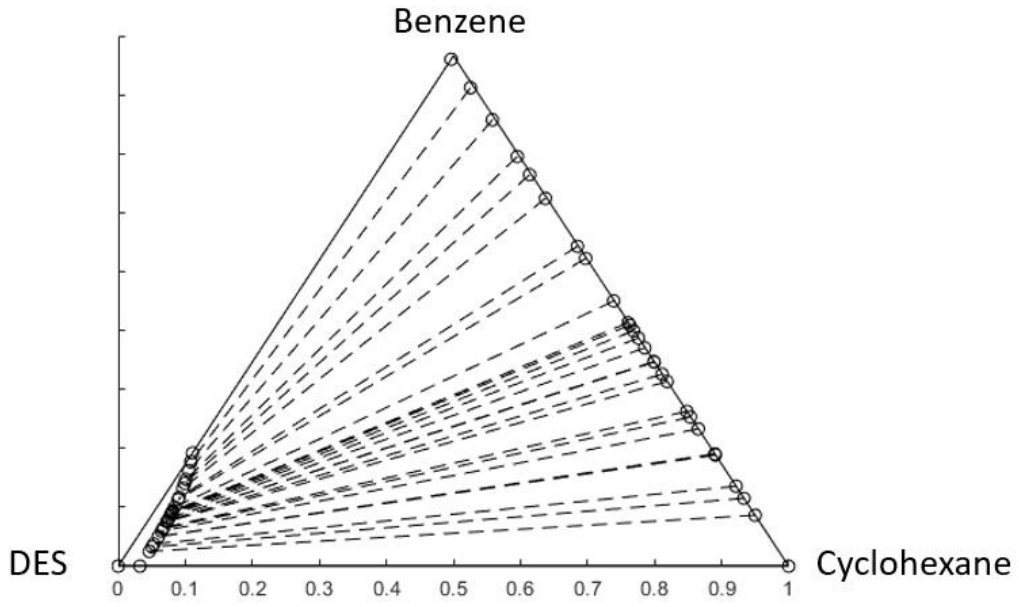


Figure 22. Predicted LLE of METPB:Gly (1:3), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

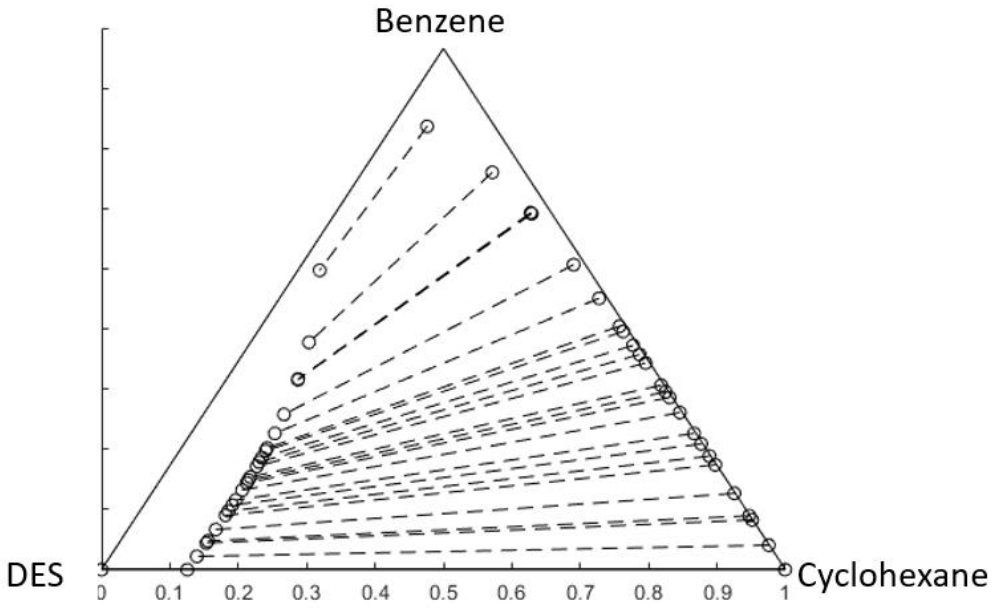


Figure 23. Predicted LLE of N_{6666}Br:Gly (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

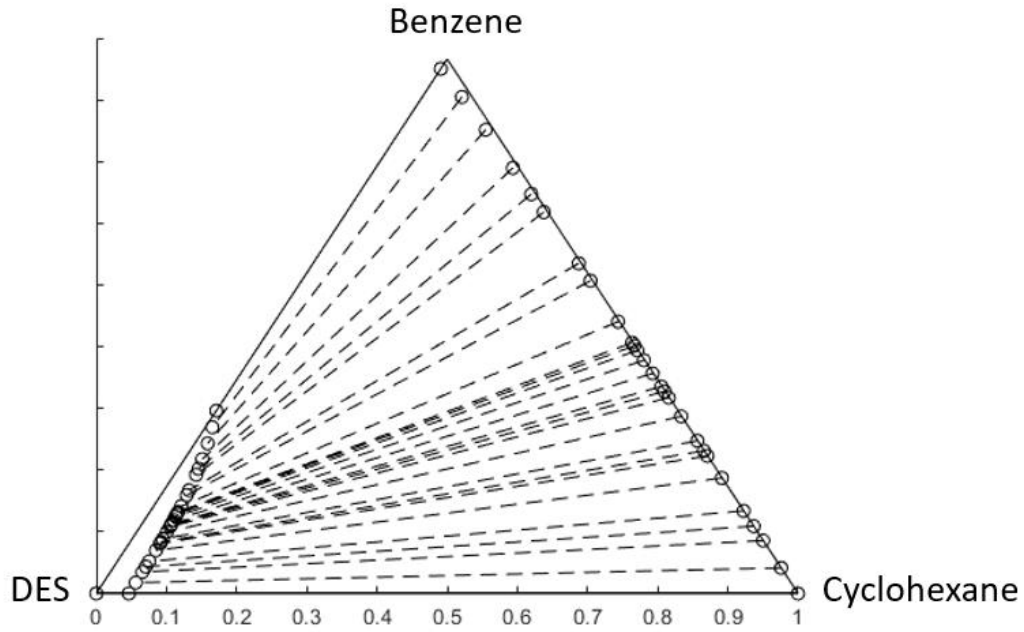


Figure 24. Predicted LLE of METPB:EG (1:3), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

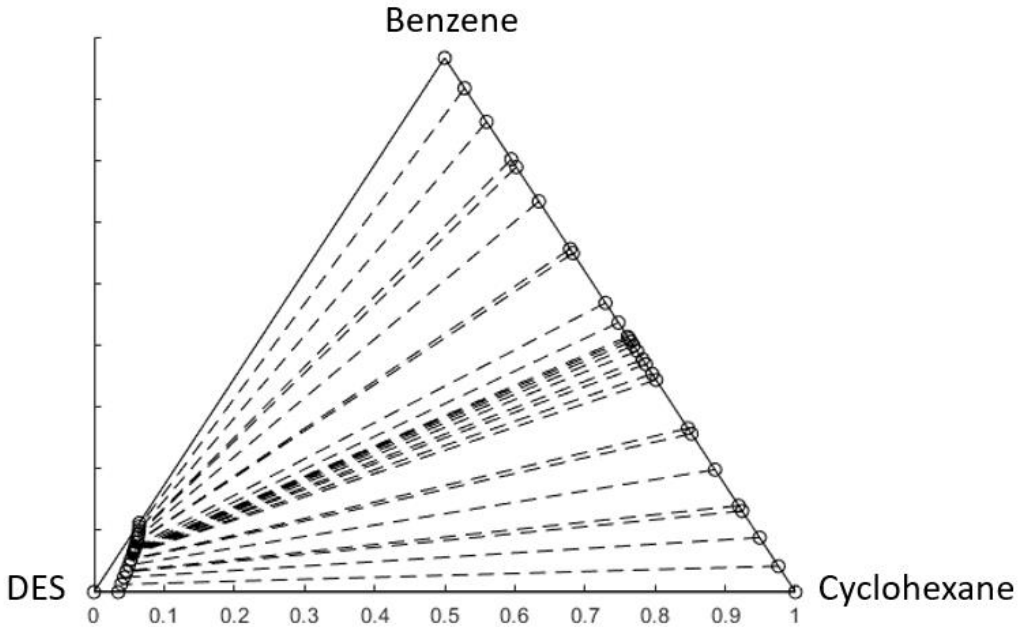


Figure 25. Predicted LLE of ChCl:EG (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

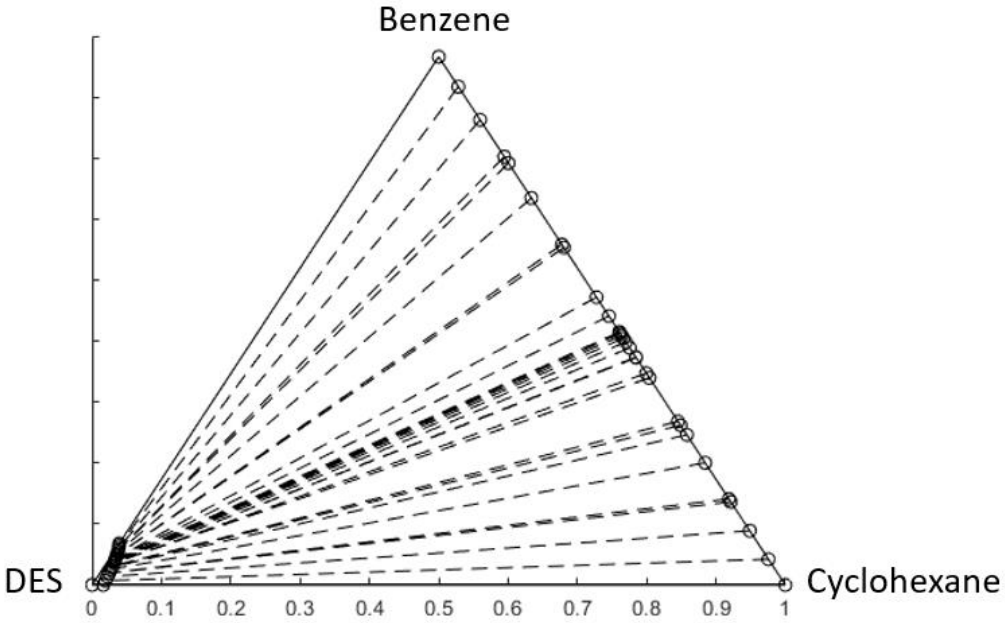


Figure 26. Predicted LLE of ChCl:Gly (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction



Figure 27. Predicted LLE of N_{4444}Br:EG (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

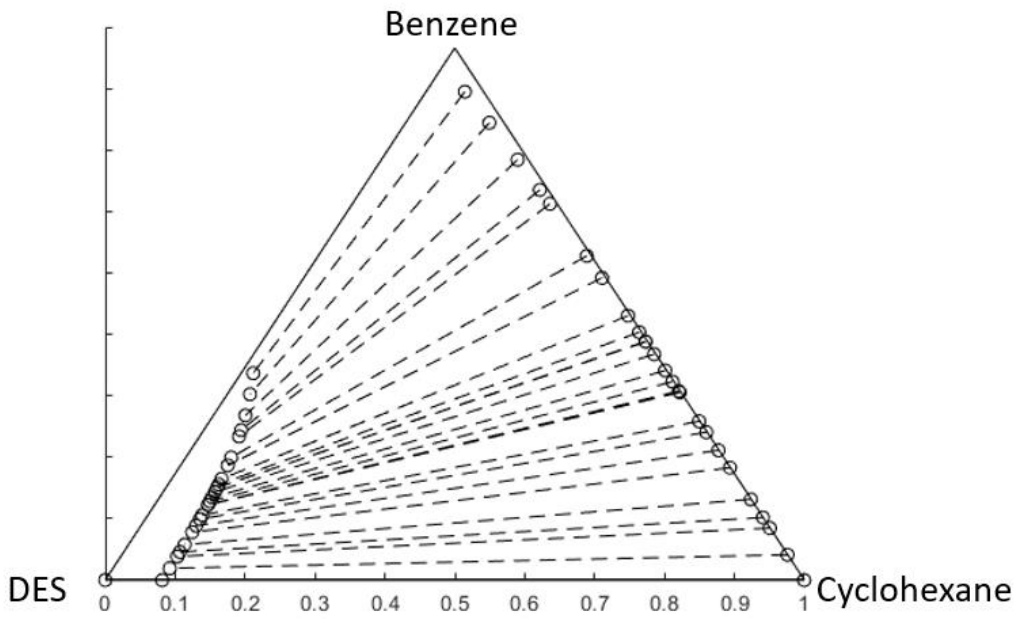


Figure 28. Predicted LLE of N_{4444}Br:Gly (1:2), Cyclohexane and Benzene at 25°C. Concentrations in Mass Fraction

Chapter 3

3.1 Solvatochromic Dye Measurements, Polarity, and Polarizability

Experimental measurements of DES properties are often used to predict how the solvents will perform in solute extraction. One of these properties is polarizability, which is distinctly different from polarity. While the electron density profiles discussed in Chapter 2 show the relative charge of the surface of the DES, polarizability describes how these charges can be shifted by outside influences. Water is both polar and polarizable, while cyclohexane is non-polar and non-polarizable. These properties are not mutually tied together though, since polarity is based and symmetry without an outside influence, and polarizability is how electron distribution is changed by an outside influence. This is particularly of interest when considering benzene. Benzene has a symmetric distribution of electrons when considering a plane along the aromatic carbon backbone, which means it is not polar. However, polarity can be induced by an outside influence that shifts electron density to one of the delocalized π-orbitals. Figure 29 shows this electron distribution on the surface of benzene.

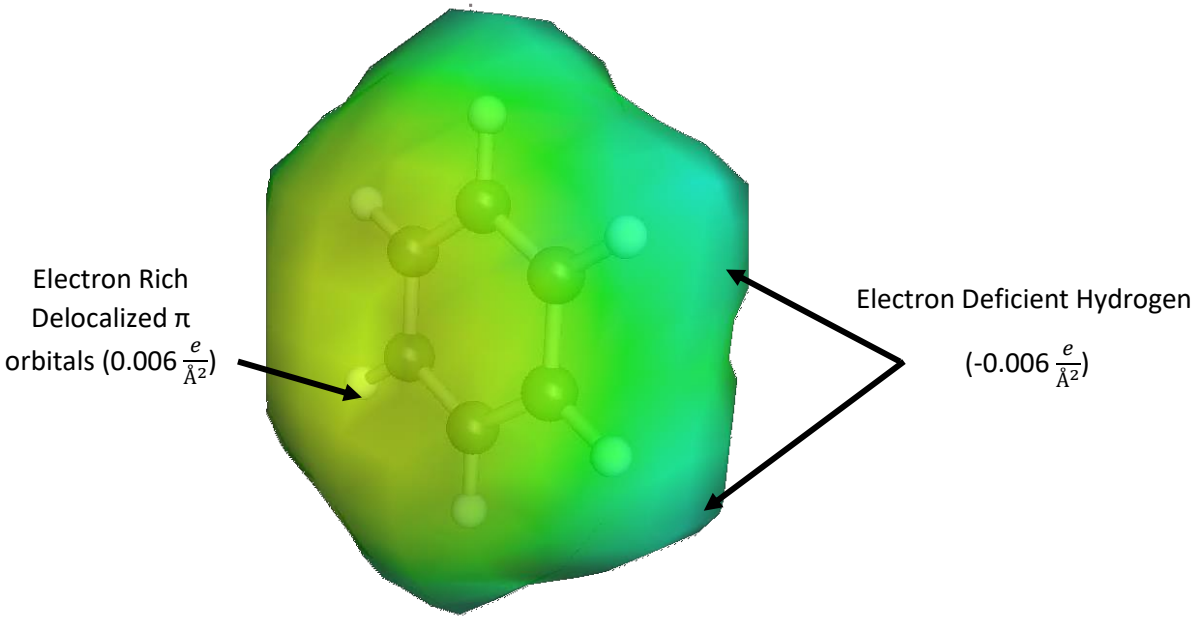


Figure 29. Benzene Surface Electron Density. Bluer shading denotes electron deficient regions (around hydrogens) while yellow shading denotes electron rich regions (around carbon's π orbitals)

DES can be influenced in the same way, and this effect can help dissolve species that favorably interact with polar surfaces, such as the polarizable aromatics of benzene, toluene, and xylene. Measuring polarizability is significantly more difficult than polarity however, as it requires the introduction of a measurement device at a molecular level. To this end solute dyes, such as functionalized 4-nitroaniline dyes, are used. These dyes have a solvatochromic effect, where their ultraviolet and visible light spectra change depending on the influence of their environment on their different functional groups. Strong interactions from the environment can cause solvatochromic dyes to have shorter or longer wavelength spectra. Relating the strength of this environmental effect to molecular properties such as polarity or

dipolarity through empirical relationships can then allow for measurements using these dyes. [26,31,37]

Solvatochromic dyes, or dyes that change color depending in the solvent they are in, are useful for making measurements of the properties of DES.[26,31,37] Molecular level properties such as polarity and dipolarity are most commonly described by Kamlet Taft parameters, which describe various polar properties of molecules empirically. Equation 2 below shows the generalized form of Kamlet-Taft solvatochromic parameters.

$$XYZ = XYZ_o + a\alpha + b\beta + s\pi^* \qquad \text{Eq 2.}$$

Where XYZ is some polarity related property, XYZ_o is the same property referenced in a nonpolar medium, α is hydrogen bonding donating ability, β is hydrogen bonding accepting ability, π^* is polarizability, and a, b, and s are all constants.[38,39] This equation essentially states that by understanding how a property is affected by polarity, changes in polar characteristics can have a quantified, predictable impact on the described property.

Previously it was discussed how polar interactions only account for a small portion of the favorable interactions that contribute to benzene solubility.[56] Describing polarity characteristics in multiple terms is more useful than simply using the strength of the polar dipole as these terms can be related to a specific polarity related

property in a more flexible manner (i.e. considering how dipole sensitivity affects solubility of a solute rather than just examining the strength of the dipole). Kamlet-Taft parameters may therefore be useful in selecting DES that have polar characteristic favorable for benzene solubility by better describing the multiple forces that contribute to benzene solubility.

Florindo et al. have demonstrated through experimental measurement of Kamlet-Taft parameters that α and β are generally consistent within specific DES families.[26] For example, choline chloride paired with ethylene glycol will have nearly the same hydrogen bond accepting and donating parameters (α and β) as choline chloride and glycerol (specifically at the eutectic ratio).[26] Cai et al. reported similar results with respect to choline chloride based DES, were even significant changes in HBD functionality resulted in little change to the Kamlet-Taft parameters.[74] This consistency in Kamlet-Taft parameters was also observed in quaternary ammonium based DES when paired with carboxylic acids ranging from butanoic to decanoic acids, even those this spans very hydrophobic to very hydrophilic characteristics.[37] Given Equation 2 and knowing that α and β are generally consistent amongst similar DES, making an informed selection of DES for BTX solubility should consider solvents with π^* parameters matching solvents that have a high solubility for BTX. In other words, if a DES has similar Kamlet-Taft parameters to solvents such as sulfolane, ethylene glycol, and glycerol it is probably a good solvent for BTX

extraction. To evaluate this hypothesis the Kamlet-Taft parameters and BTX solubility of the selected solvents were compared to the reference solvents.

3.2 Solvatochromic Dye Measurements Methodologies

Preparation of a solvatochromic dye was done with N,N, diethyl 4-nitroaniline (98%, Combi-Blocks) and acetone (99.5%, Sigma-Aldrich). The dye was first tested by direct addition to several reference solvents including; glycerol (99% Acros Organics), ethylene glycol (99%, Acros Organics), DI water, cyclohexane (99.9%, Fisher Scientific), Dimethyl sulfoxide (99.9%, Fisher Scientific), and acetone. The dye was then retested by a solvent delivery method to expedite the experimental process by first preparing a solution of $1.4x10^{-5}$ M dye in acetone and then evaporating the solvent to deliver the dye directly to a cuvette. Samples had a concentration of $2x10^{-5}$ to $2x10^{-7}$M when measured by the UV-Vis. All UV-Vis measurements were taken with a DR6000 Hach UV-Vis spectrometer using a quartz cuvette rated for 190nm-1100nm (Fischer Catalog #14-958-128). These measurements were within 0.01 abs of the direct addition measurements. The solvent delivery method was used for all measurements presented in this study. Additionally, the Beer-Lambert law was evaluated in reference solvents (glycerol, ethylene glycol, DI water, cyclohexane) to ensure that dye concentration was sufficiently low as to not interact with itself. Equation 3 shows the Beer-Lambert law where A is absorbance, ε is the molar absorbance coefficient ($M^{-1}cm^{-1}$), C is molar

concentration (M), and b is the optical pathlength (cm). The dye had an R^2 value of 0.999 for all reference solvents when evaluated at concentrations of $2x10^{-5}$ to $2x10^{-7}$ M mol/L of dye concentration in each solvent.

$$A = \varepsilon bC \qquad \text{Eq. 3}$$

DES were prepared to be evaluated by the solvatochromic dye. The following chemicals were used for the DES; glycerol (99% Acros Organics), ethylene glycol (99%, Acros Organics), Choline Chloride (99%, Acros Organics), methyltriphenylphosphonium bromide (98%, TCI), tetrahexylammonium bromide (98%, Alfa Aesar), and tetrabutylammonium bromide (98%, Alfa Aesar) . The DES were prepared according to the ratios from Table 1. DES components were first combined at the specified ratios in a sealed flask at room temperature and then heated to 50°C on a heating plate with gentle mixing. DES were allowed to cool to room temperature (20°C) under natural cooling and were sealed with Parafilm to prevent any moisture uptake from the environment.

Dipolarity was calculated from Equation 4, where $\overline{V}_i$ is the wavenumber of either the solvent, cyclohexane, or dimethyl sulfoxide (DMSO) as noted.[26] In this scale the solvent dipolarity is normalized between cyclohexane and DMSO at 0 and 1 respectively.

$$\pi^* = \frac{\overline{V}_{Solvent} - \overline{V}_{DMSO}}{\overline{V}_{Cyclohexane} - \overline{V}_{DMSO}} \qquad \text{Eq 4.}$$

3.3 Mass Transfer Kinetics and Equilibrium Measurement Methodologies

Both experimentally determined and model predicted liquid-liquid equilibrium between DES, cyclohexane, and BTX can be used to help select DES. However, this is only one factor to consider when designing naphtha purification systems. Mass transfer kinetics are another important factor that, unlike properties such as density and viscosity, can be very difficult to predict with models. Thus, mass transfer kinetics were measured experimentally to complement existing literature values for DES properties as well as those predicted by the COSMO models previously discussed. Additionally, these experiments were used to verify the predicted equilibrium conditions. With the overall goal of estimating DES performance in a full BTX removal process the measurement of kinetic parameters will help size the purification units and give ranges of kinetic parameters compared to common references such as the HBD components of the DES. If mass transfer kinetics in DES are similar, or at least predictably related, to the HBD's parameters in the same way the polarity properties like the Kamlet Taft parameters are, entire families of DES can be prioritized for evaluation. In other words, screening for DES components can be expediated by evaluating first the candidate HBD components and then DES made from those HBD's that have favorable performance for BTX removal.

Mass transfer kinetics were measured for benzene from cyclohexane into select DES; METPB:EG (1:3), METPB:Glycerol (1:3), N_{4444}Br:EG(1:2), N_{4444}Br:Glycerol (1:2). Benzene was selected as it accounts for over 95 mass% of aromatics in light naphtha.[75] Initial concentrations of benzene ranged from 2-10 mass% with a solvent:solute ratio reflecting those typically used in sulfolane based processes (1:1.5).[6] DES and model naphtha made of benzene and cyclohexane were mixed in a sealed flask with a constant rate of stirring at 25°C in a 65mm diameter flask with a 25mm magnetic stir bar. Stirring rate was varied from 400-600 RPM. The organic phase was regularly sampled and analyzed using a DR6000 Hach UV-Vis spectrometer with a quartz cuvette rated for 190nm-1100nm. A Beer-Lambert law calibration at 238.8 nm was made using benzene in cyclohexane ranging from $3.5x10^{-3}$ to 0.1 mass% benzene, which is shown below in Figure 30. Benzene has 5 distinct peaks in the ultraviolet range at 238.8, 244.0, 249.2, 255, and 261.5 nm. 238.8 nm was chosen due to it having the best linear fit over the selected concentration range ($3.5x10^{-3}$ to 0.1 mass%). Figure 31 shows all the benzene peaks evaluated for the Beer-Lambert relation. The samples taken from the mixing experiment were diluted to be in this range. Sampling was done at 5-minute intervals for 60 minutes for determining mass transfer kinetics. Samples were then taken at 12-hour intervals until concentration of the benzene was constant to find concentration at equilibrium.

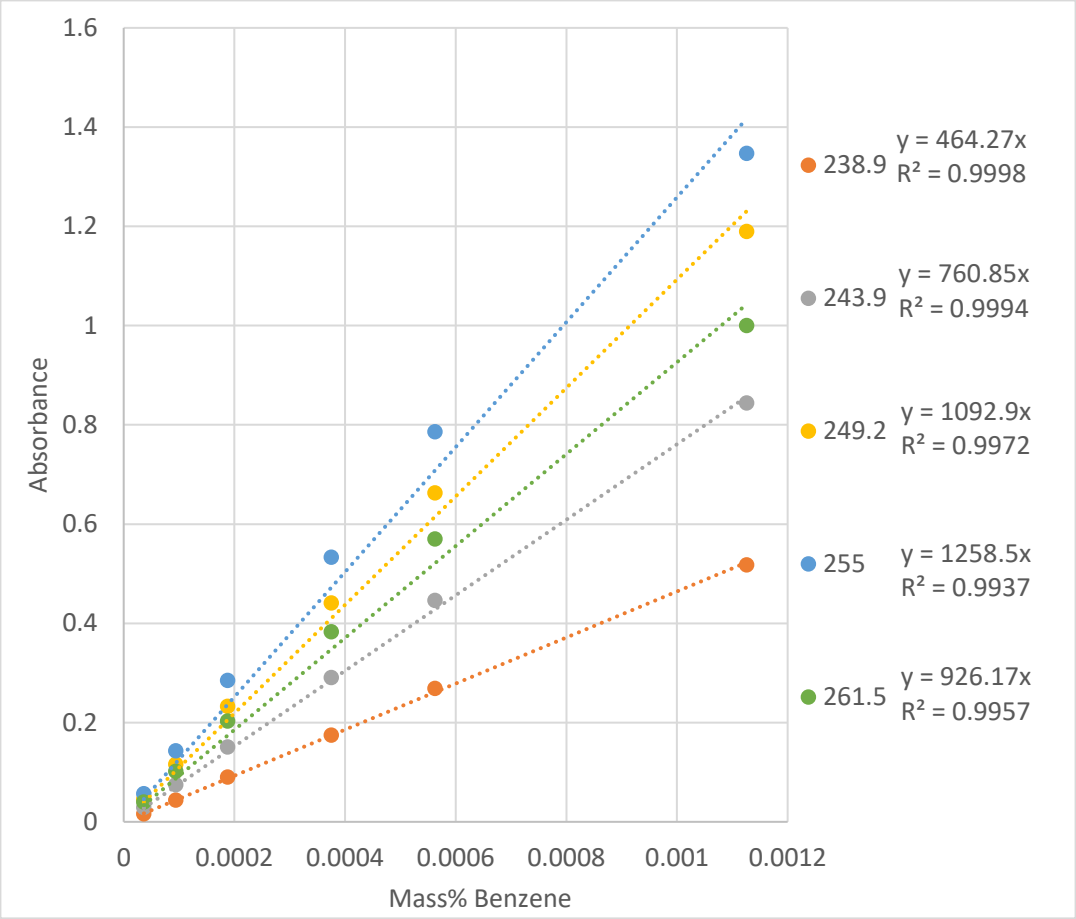


Figure 30. Beer-Lambert Relation for Benzene in Cyclohexane at Notable Absorbance Peaks

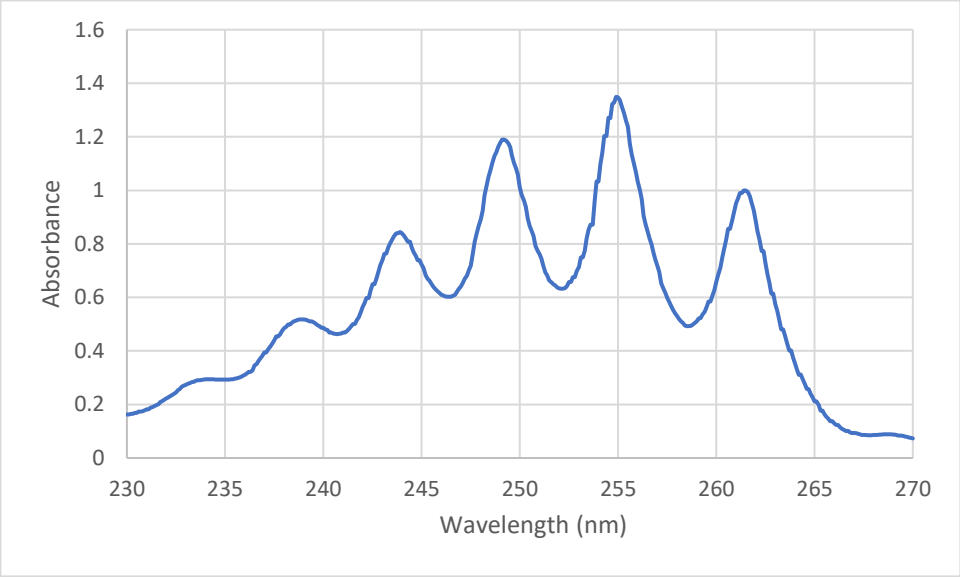


Figure 31. UV-Vis Spectra for Benzene in Cyclohexane

The equilibrium concentration of the DES-cyclohexane-benzene system was modeled in COSMOtherm using the same process as described in Chapter 2. The predicted concentration of benzene was compared to the measured concentration to evaluate the accuracy of the COSMO Model. The absorbance measured experimentally was converted into mass% benzene removal by Equation 5.

$$\%\ Benzene\ Removal = 1 - A_t/A_o \qquad \text{Eq. 5}$$

Where A_t is the absorbance at sampling time t and A_0 is the absorbance of the initial cyclohexane benzene solution. To ensure that the DES components were not affecting absorbance in the ultraviolet range, a blank test of each DES with cyclohexane was completed. None of the DES or individual HBD's had an effect on the cyclohexane absorbance at 238.8nm.

3.4 Solvatochromic Dye Measurement, Equilibrium and Kinetic Results

Table 5 shows the polarizability of both DES and HBD on a relative scale where cyclohexane and DMSO are fixed at 0 and 1 respectively. The DES all have very similar polarizability with values ranging from approximately 1.10 to 1.20. All glycerol-based DES have polarizability similar glycerol itself. METPB:EG has a slightly higher value than ethylene glycol at 1.22±0.02 and 1.02±0.02 respectively. N_{4444}Br:EG and ChCl:EG are also higher than ethylene glycol at 1.09±0.02 and 1.10±0.04 respectively. N_{4444}Br:Gly has a high polarizability given its HBD. DES made from N_{4444}Br typically have π^* values ranging from 0.98-1.08, with the highest values occurring at the eutectic point.[76] Unlike N_{4444}Br DES which have relatively consistent polarizability, ChCl containing DES have been shown to have polarizability that varies significantly with different HBD's from π^* values of 0.82 for ammonium thiocyanate to 1.15 with ethylene glycol as measured here.[32]

Table 5. Polarizability of DES and Reference Solvents

Solvent	Relative Polarizability (π^*)
Cyclohexane	0
DMSO	1
Ethylene Glycol	1.02±0.02
Glycerol	1.15±0.02
METPB:EG (1:3)	1.22±0.03
N_{4444}Br:EG (1:2)	1.09±0.02
ChCl:EG (1:2)	1.10±0.04
ChCl:Glycerol (1:2)	1.15±0.02
METPB:Glycerol(1:3)	1.15±0.02
N_{4444}Br:Glycerol (1:2)	1.18±0.02

Given the large range of benzene solubility amongst the selected DES (from 0.08g benzene/g solvent for ChCl:Glycerol to full miscibility for N_{4444}Br:EG), which is more heavily influenced by nonpolar solvent characteristics than polar, it is evident that the non-polar characteristics are the main reason for the differences in BTX solubility in the solvents.[56] In regards to predicting DES polar properties from component properties, for these DES the polar properties are very similar to those of the HBD for glycerol while the ethylene glycol based DES are consistently more polarizable.

The equilibrium concentration of benzene in cyclohexane was both experimentally measured and predicted with COSMOtherm. The initial concentration of benzene was between 2 and 10 mass% with a DES:organic phase mass ratio of 1:1.5. Overall the COSMO predictions for ethylene glycol based DES were very accurate, with root-mean-square deviations (RMSD) below 1% for both N_{4444}Br:EG and METPB:EG. The glycerol systems were less accurately modeled by COSMO, with RMSD's of 4% for N_{4444}Br:Glycerol and 6% for METPB:Glycerol. At the specified solvent:solute ratio N_{4444}Br:EG removed the most benzene at 30±1%. METPB:EG and N_{4444}Br:Glycerol performed equally removing 19±2 and 20±2% respectively. METPB:Glycerol was the worst performing DES only removing 9±1% of the benzene from the organic phase. METPB DES performed worse than their N_{4444}Br counterparts most likely due to the lack of neutrally charged surface, as shown in Figure 16 and Figure 17.

All solvents performed consistently for initial benzene concentrations below 10%, with small variations within 1% relative benzene removal%. For the purpose of estimating the removal of benzene from cyclohexane it can be assumed that the benzene removal amount is constant at the specified solvent:solute ratio of 1:1.5. Cyclohexane leaching was also consistent at initial cyclohexane mass%'s varying from 90 to 99%. COSMO predicted cyclohexane leaching for N_{4444}Br:EG and N_{4444}Br:Glycerol was 9 mass% of the initial cyclohexane. METPB:EG and METPB:Glycerol only leached 2 and 3 mass% of the initial cyclohexane

respectively. Under the same conditions pure glycerol and ethylene glycol were able to remove 11 and 16 mass% benzene while leaching 2 and 3 mass% of the initial cyclohexane respectively.

Mass transfer kinetics were fit to a simple first order equation as shown below in Equation 4.

$$e^{-kt} = \frac{C \quad - C^e}{C^i - C^e} \qquad \text{Eq. 4}$$

Where k is the time coefficient (min^{-1}), t is time in minutes, C is the concentration, C^e is the equilibrium concentration, and C^i is the initial concentration at a time of 0 minutes. All concentrations refer to benzene mass% in cyclohexane. Figure 32 shows a sample of this data at an initial concentration of 2 mass% benzene.

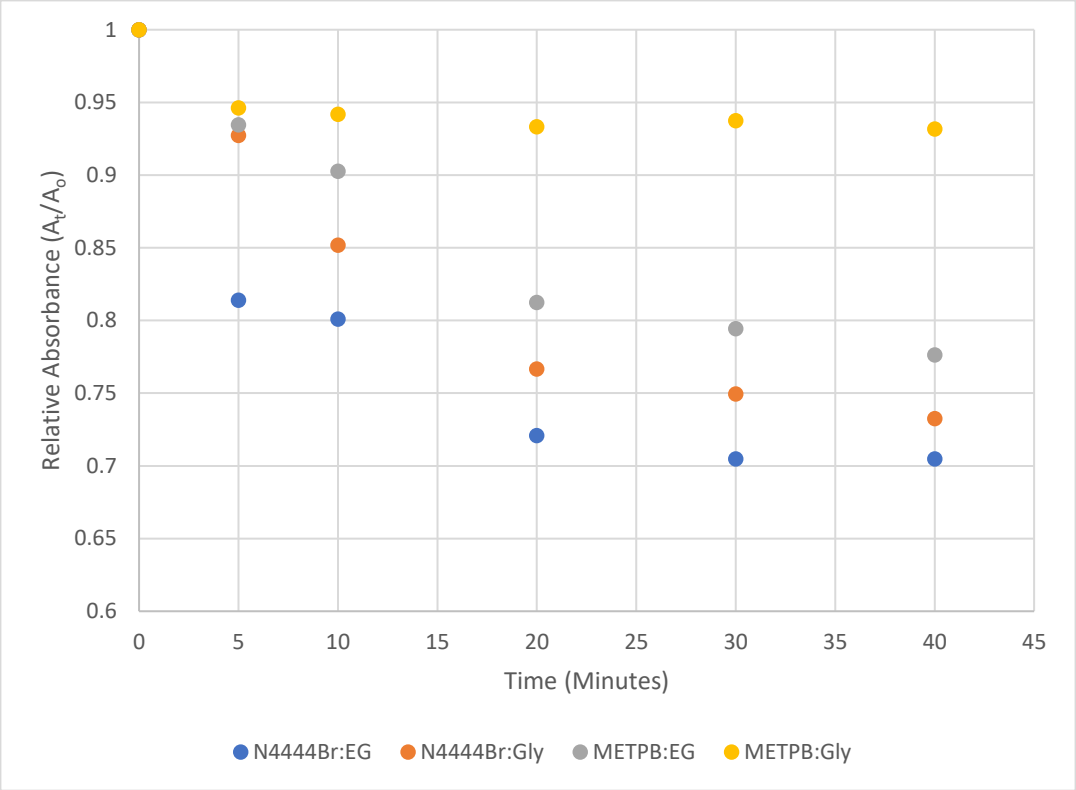


Figure 32. Benzene Concentration in Cyclohexane; Sample Kinetics for Initial Concentration of 2 Mass% Benzene

Solving Equation 4 for various measured concentration at known times will give the time coefficient k. Solving where e^{-kt}=0.5 will give the half-life time, or the time it takes for the solution to move halfway to equilibrium. These times are useful for comparing the relative speed at which each DES removes benzene from cyclohexane. These values are shown in Table 6. N_{4444}Br:EG and METPB:Gly had the shortest half-life times at 3.5 and 3.6 minutes respective. N_{4444}Br:Gly and METPB:EG were far slower at removing benzene with longer half-life times of 7.5 and 8.1 minutes respectively.

Table 6. Time Coefficient and Half-Life Times for Benzene Transfer into DES from Cyclohexane at 25°C

DES	k(min^{-1})	t(1/2)
N_{4444}Br:EG (1:2)	0.199	3.5
N_{4444}Br:Gly (1:2)	0.092	7.5
METPB:EG (1:3)	0.086	8.1
METPB:Gly (1:3)	0.192	3.6

N_{4444}Br:EG has both the highest amount of benzene removed and the lowest half-life time, however it also has a high amount of cyclohexane uptake. METPB based DES remove less benzene but also have a far lower uptake of cyclohexane. Interestingly METPB:Gly reaches equilibrium far faster then METPB:EG, while N_{4444}Br has the opposite with N_{4444}Br:EG having faster transfer than N_{4444}Br:Gly. This suggests that mass transfer is not dependent solely on the HBA or the HBD, but rather the DES. This makes predicting DES performance from just HBA or HBD properties more difficult, but also justifies the continued use of experimental methodologies for evaluating DES even if equilibrium conditions can be accurately estimated by modeling methods such as the COSMO method used in this work.

Chapter 4

4.1 Benzene Removal from Naphtha Model Methodologies

Liquid-liquid extraction of benzene from cyclohexane was modeled by using the equilibrium and kinetics parameters that were determined from Chapter 3. These parameters consistent of liquid-liquid equilibrium data for cyclohexane, benzene, and DES that were first predicted computationally and then confirmed experimentally as well as mass transfer kinetics parameters for benzene transfer into he DES phase that were measured. The efficiency and effectiveness of this removal process will be used to evaluate how each DES, selected from candidates evaluated for benzene and cyclohexane solubility, performs compared to each other and sulfolane (the industry standard for aromatic removal from alkanes). DES pairs made of METPB:Ethylene Glycol, METPB:Glycerol, N_{4444}Br:Ethylene Glycol and N_{4444}Br:Glycerol were selected for evaluation. Though light naphtha can contain other aromatics such as toluene and xylene, only benzene is being considered in this model as it makes up the vast majority of the aromatics in naphtha. The objective of the model process is to purify a stream of 1,000 kg/hr of model naphtha that contains 10 mass% benzene using a separation unit that operates near room temperature at 25°C.[75] The extraction process was based off of the standard that is used for sulfolane systems, as shown in Figure 1. This process first mixes the aromatic rich naphtha

stream with the solvent and then regenerates the solvent and purifies the naphtha product separately.

Figure 33 shows the process flow diagram for benzene removal, where mixing and phase separation units are connected in series to purify the naphtha stream. Fresh DES is introduced into each mixing unit where benzene moves from the naphtha phase into the solvent phase. Stream naming convention depends on the unit number in the process where N denotes the unit number (from 1 to 7). For example, the 5th mixing and separation unit with have stream numbers 5-1 for the naphtha inlet while the final, 7th flash separation stage will have stream numbers 7-6 for the benzene product.

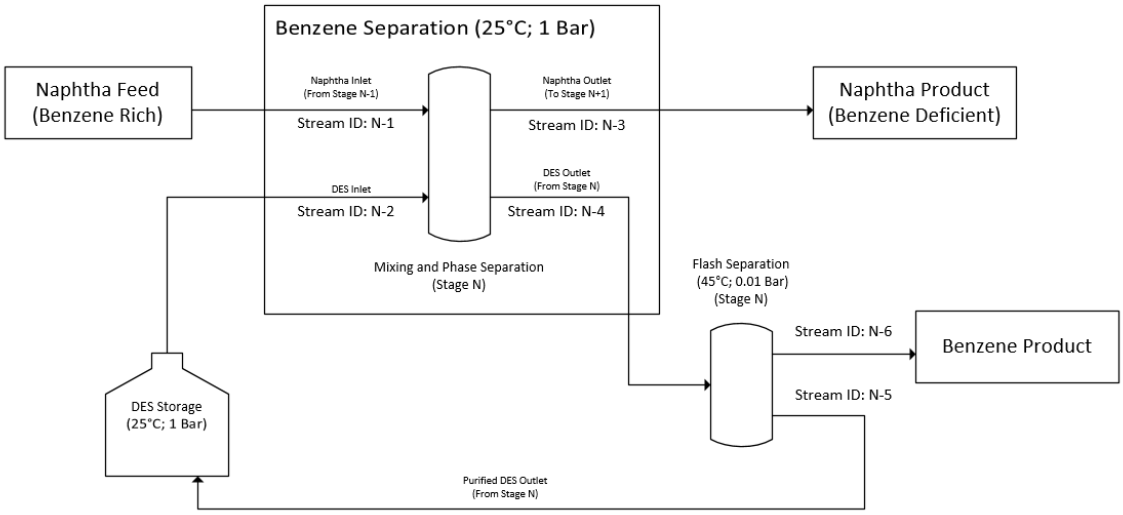


Figure 33. Simplified Process Flow Diagram for Benzene Removal via DES

The amount of benzene removes follows the equilibrium and kinetics data from Chapter 3, given by Equation 4.

$$e^{-kt} = \frac{C \quad - C^e}{C^i - C^e} \qquad \text{Eq. 4}$$

Where k is the time coefficient (min^{-1}), t is time in minutes, C is the concentration, C^e is the equilibrium concentration, and C^i is the initial concentration at a time of 0 minutes. All concentrations refer to benzene mass% in cyclohexane. All k time coefficient values are taken from Table 6. The product of each phase separation unit is sent to the next mixing unit, where more fresh DES is introduced. A total of 7 mixing and phase separation pairs were used, which gives a total solvent:naphtha feed ratio of 4.67:1 (each individual unit is 1:1.5), comparable to similar sulfolane units that have ratios of 4-6:1.[6] Each solvent had residence times in the mixing units equal to 2 half-life times, essentially allowing for the mixture to achieve 75% of the equilibrium value. This is a very conservative estimate of mass transfer kinetics, but does reflect some of the difficulties of maintaining well mixed units with DES (high density and high viscosity) that are not considered elsewhere in this model. The aromatic rich DES stream exiting the phase separation unit is sent to a vacuum flash distillation unit where the low pressure is used to remove cyclohexane and benzene from the DES. These units operate at 45°C and 0.01 bar absolute. This removal is determined using Henry's law, where Henry's law constants were determined using COSMO as described in Chapter 2. Table 7 below shows Henry's law constants for cyclohexane, benzene, and HBDs The partial pressure of the HBD in the DES phase

is far lower than if they were pure. This is consistent with experimental data that has determined the vapor pressure of HBD in DES at the eutectic point is usually negligible with vapor pressures 100-10,000 times lower than pure HBD's.[77]

Table 7. Henry's Law Constants and Heat of Solvation for Solutes in DES

Henry's Law Constant (bar/mole fraction); 25°C

DES	Benzene	Cyclohexane	HBD
N4444Br:EG	0.35	1.97	7.0E-05
N4444Br:Gly	0.42	2.41	1.7E-06
METPB:EG	0.50	4.54	1.0E-04
METPB:Glycerol	0.61	5.53	3.0E-06

Henry's Law Constant (bar/mole fraction); 45°C

DES	Benzene	Cyclohexane	HBD
N4444Br:EG	0.74	3.99	4.9E-04
N4444Br:Gly	0.88	4.76	1.8E-05
METPB:EG	1.08	8.83	7.0E-04
METPB:Glycerol	1.26	10.36	2.9E-05

ΔG Solvation(kJ/mol); 45°C

DES	Benzene	Cyclohexane	HBD
N4444Br:EG	-16.11	-11.81	-37.22
N4444Br:Gly	-15.41	-11.09	-46.20
METPB:EG	-15.67	-10.23	-36.71
METPB:Glycerol	-14.91	-9.43	-45.18

Partial pressure of each volatile species was calculated by Equation 7, using Henry's law.

$$P_i = X_i H_i \qquad \text{Eq. 7}$$

Where Pi is partial pressure in bars, Xi is mole fraction in the liquid, solvent phase, and Hi is the Henry's law constant in $\frac{bars}{mole\ fraction}$. For this specifical calculation, since the Henry law constant are determined using COSMO, mole fraction considers all ionic species separately. Therefore, HBA's are considered as two separate moles for this calculation: one anion and one cation. Change in liquid mole fraction of volatile species as pressure changes was determined by the total differential of pressure using Equation 8 and the sum of partial pressures in Equation 9.

$$dP = \sum_i^n \frac{\partial P}{\partial X_i} dX_i \qquad \text{Eq. 8}$$

$$P \quad = \sum_i^n X_i H_i \qquad \text{Eq. 9}$$

Energy requirement for the vacuum flash unit was determined in several individual parts. Energy required for moving solutes into the gas phase was determined by COSMO calculation and are listed in Table 7. Energy required for maintaining the vacuum flash units was based off of industrial rotary vane vacuum pumps and is set at 12 $\frac{W}{m^3 hr}$.[78] Volumetric flow rate of the gaseous product stream from the vacuum flash units was modeled in ASPEN Plus using the default Flash unit operation with the Redlich-Kwong equation of state model.

Sensible heat required for heating the solvent stream from 25 to 45°C was determined by using the heat capacities for each individual species. Table 8 shows the heat

capacity of species, which is assumed to be constant. METPB DES values are taken from literature while $N_{4444}Br$ DES are from the heat capacity of the individual components. These values are assumed to be constant over this small of a temperature change.

Table 8. Specific Heat Capacity for DES, Cyclohexane, and Benzene

Species	Specific Heat Capacity (kJ/kg°C)
$N_{4444}Br$:EG	1.42[79]
$N_{4444}Br$:Gly	1.54[79]
METPB:EG	1.84[80]
METPB:Gly	2.09[81]
Cyclohexane	1.71
Benzene	1.25

To repressurize the DES after regeneration in the vacuum flash units a pump is required. Total work required to achieve this pressure change, W_{pump} is calculated from Equation 10, where $\dot{v}$ is the volumetric flow rate in m^3/hr, ΔP is the pressure differential in pascals, and F is a pressure drop factor to account for losses in the rest of the system. F is assumed to be 3.

$$W_{pump} = F\dot{v}\Delta P \qquad \text{Eq. 10}$$